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Md. Nahin Alam

AI Automation Engineer & Researcher

Professional Summary

Results-driven **AI Automation Engineer** and **Graduate Researcher** with proven expertise in orchestrating production-ready LLM ecosystems, scalable RAG pipelines, multi-agent systems, and spatial/visual computing workflows. First-author of peer-reviewed AI publications across venues like ICCIT and ICMLAS. Experienced in deploying cloud-native machine learning pipelines on AWS/GCP, cutting data lookup/processing cycles by **60%**, and delivering reliable, high-performance web applications.

Technical Skills

AI / ML Core	**PyTorch**, TensorFlow, Hugging Face, Scikit-learn, XGBoost, LightGBM, FastAI
LLMOps & RAG	**RAG Architecture**, FAISS, Google Genkit, Semantic Search, Multi-Agent Systems, n8n
Vision & Multimodal	**CLIP**, ViLT, BLIP, BioViL-T, Vision-Language Models (VLMs), Radiology AI
Cloud & Backend	**FastAPI**, Node.js, AWS, GCP, **Docker**, Kubernetes, GitHub Actions (CI/CD), PostgreSQL
Frontend & Web	**Next.js 15**, React 19, TypeScript, Tailwind CSS v4, Shadcn UI, Prisma ORM, TanStack Query
Languages	**Python**, TypeScript, C/C++, Java, SQL / NoSQL, MATLAB

Experience

Jan 2026 – Present **AI & Automation Engineer**, *MySoftHaven (BD) Ltd.*, Dhaka, Bangladesh
- Architect production-grade AI automation solutions for enterprise and government-level applications.
- Deploy multi-tenant SaaS ecosystems incorporating RAG chatbots (**FAISS + Gemini 1.5 Flash**) and AI lead orchestrators, reducing manual data processing time by **60%+**.
- Implement automated CRM synchronization workflows connecting Salesforce and HubSpot via custom API middleware.

Jan 2026 – Present **Graduate Research Assistant**, *SADIA Lab, KFUPM*, Dhahran, KSA
- Pioneer end-to-end Sign Language Generation (SLG) pipelines using spatial computing architectures and 3D pose estimation.
- Develop deep learning data alignment workflows for high-dimensional video datasets, improving synthesis realism by **35%+**.
- Co-author research manuscripts targeted at top-tier Computer Vision conferences (**CVPR, ICCV, MICCAI**).

Jan 2025 – Present **AI Research Assistant (Remote)**, *Centre for UntieAI Research, CU Boulder*, USA
- Constructed automated ETL pipelines to process over **500 GB+** of multimodal medical imaging and NLP datasets, cutting preparation time by **50%**.
- Co-authored 3 high-impact manuscripts on scalable AI methodologies under review at KDD, ACL, and SaTC.

Jun 2024 – Oct 2025 **Cloud Infrastructure & Automation Specialist**, *The Data Island*, Dhaka, Bangladesh
- Engineered containerized cloud architectures on AWS and GCP using Docker and Kubernetes, achieving **99.9% system uptime**.
- Optimized RESTful APIs and PostgreSQL database schemas, reducing query response times by **35%** to handle **5,000+ concurrent requests**.

Jan 2025 – Oct 2025 **Vision-Language Model Researcher**, *ML Lab, North South University*, Dhaka, Bangladesh
- Benchmarked state-of-the-art VLMs (CLIP, ViLT, BioViL-T) across **10,000+** data points to analyze architectural bottlenecks in medical AI.
- Built automated evaluation frameworks for *RadMultiBench*, accelerating model testing speed by **3**×.

Selected Publications & Research

2026 **First Author**, *ICMLAS 2026*, Accepted
Validating a Consensus Feature Selection Pipeline for Rheumatoid Arthritis Biomarkers: A Synthetic Data Study.

2026 **Co-Author**, *SaTC 2026*, Accepted
Harnessing Vision Language Models for Negation Sensitivity in Visual Question Answering.

2026 **Co-Author**, *SaTC 2026*, Accepted
Engineering Spatiotemporal Forecasts at Scale: Dhaka Weather with State-Of-the Art Machine Learning Models.

2025 **First Author**, *ICCIT 2025*, Accepted
From Benchmarks to Deployment: A Calibrated, SHAP-Guided, Production-Ready Pipeline for Diabetes Risk Prediction.

2025 **First Author**, *Preprint / Under Review*, Under Review
RadMultiBench: A Multimodal Benchmark Revealing the Limits of Frozen Vision-Language Models in Radiology.

Key AI Projects

MySoftHaven AI Suite — **Enterprise RAG Platform** built with FastAPI, FAISS, and Gemini 1.5 Flash. Decreased enterprise data lookup latency by over 60%.

CodeGenius Platform — **AI Code Tutor** powered by Next.js 15, Firebase, and Google Genkit (Gemini 2.5 Flash) with adaptive learning paths.

Nexus Knowledge Graph — **AI Knowledge Cartographer** converting unstructured text into interactive SVG knowledge graphs via Gemini Structured Outputs.

Duari Study Companion — **Recommendation Engine** built on Next.js 15, Prisma ORM, and Recharts analytics for personalized university mapping.

Education

2026 – Present **MSc in Computer Science**, *King Fahd University of Petroleum & Minerals (KFUPM)*, KSA
Specialization in Spatial & Visual Computing | SADIA Lab Graduate Researcher

2020 – 2024 **BSc in Computer Science & Engineering**, *North South University*, Dhaka, Bangladesh
GPA: 3.81 / 4.00 (Summa Cum Laude, Dean's List)

Teaching & Certifications

Teaching — **AI/ML Lab Instructor** at North South University (Feb 2025 – Present). Instructed 50+ students/semester in PyTorch, TensorFlow, DSA, and OOP.

Certifications — Stanford Machine Learning Specialization (Coursera), Deep Learning Specialization (Coursera), FastAI Deep Learning, GCP Fundamentals.